

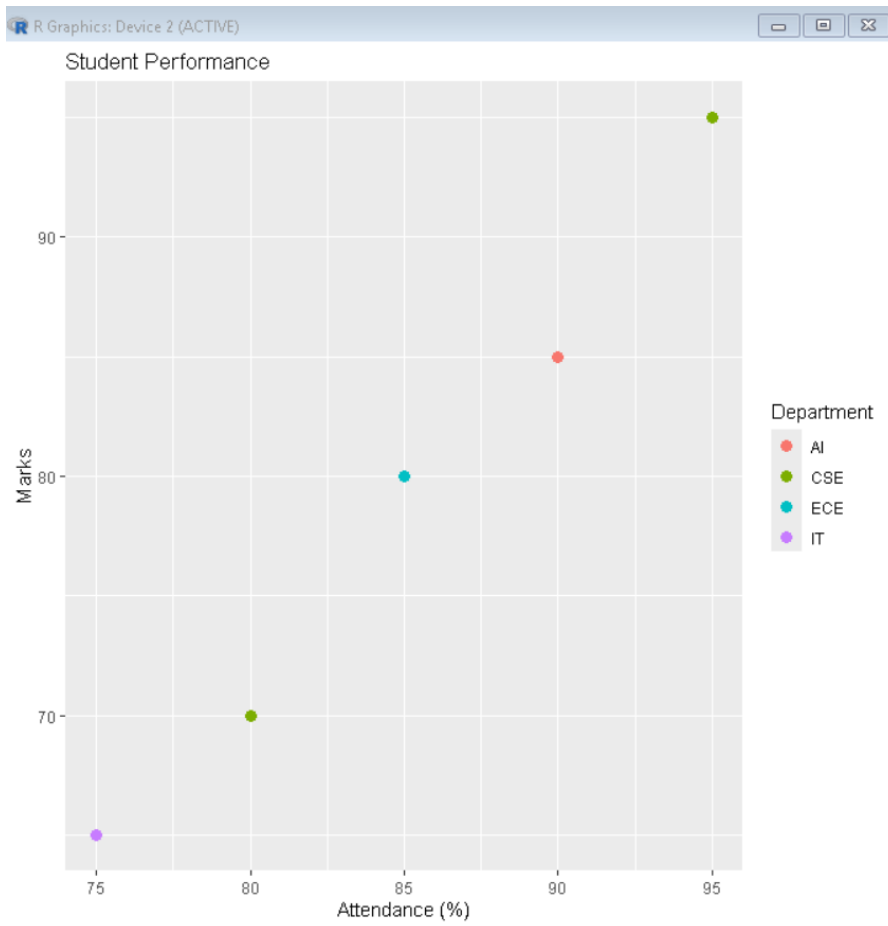
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Reg No : 192424211
Course/Code : DSA0604 - DATA HANDLING AND VISUALIZATION
Date : 10/07/26

QUESTION 1: STUDENT ACADEMIC PERFORMANCE

CODE:

```
ggplot(student, aes(x = Attendance, y = Marks, color = Department)) +  
  geom_point(size = 3) +  
  labs(  
    title = "Student Performance",  
    x = "Attendance (%)",  
    y = "Marks"  
  )
```

OUTPUT:

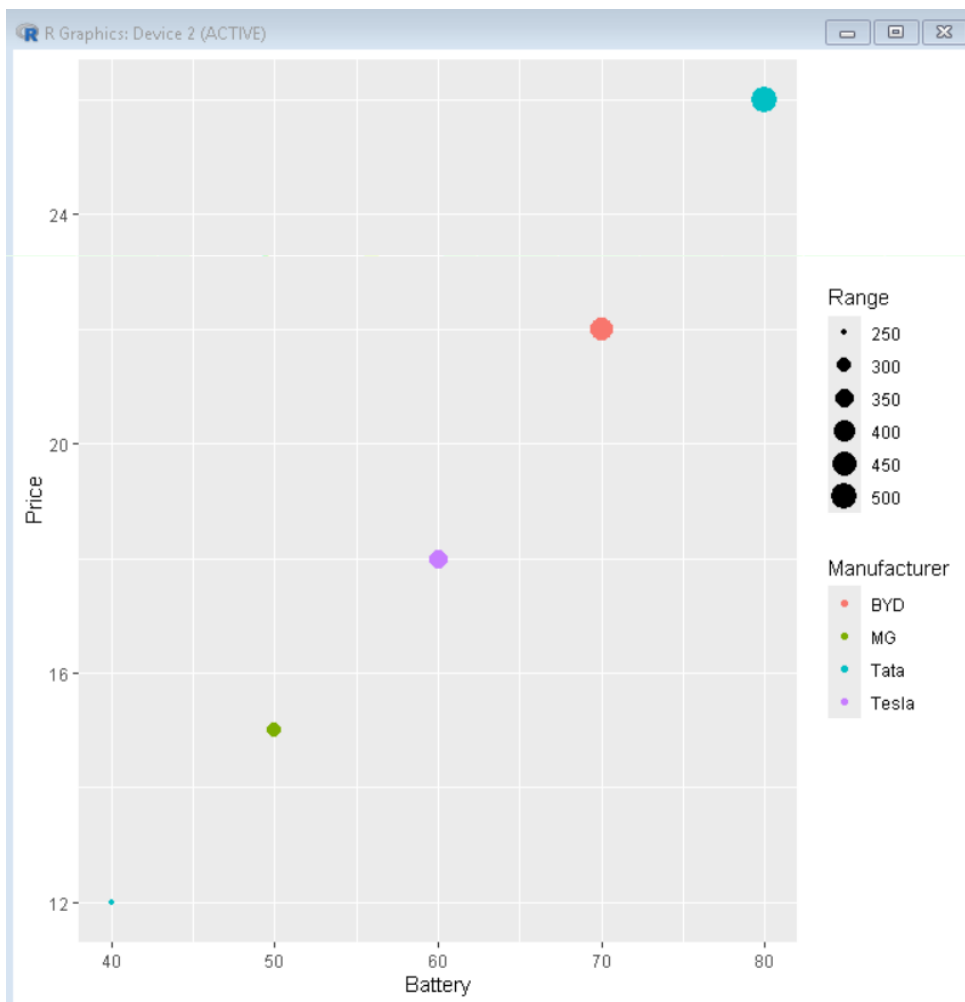


QUESTION 2: ELECTRIC VEHICLE MARKET ANALYSIS

CODE:

```
ev <- data.frame(  
  Battery = c(40,50,60,70,80),  
  Price = c(12,15,18,22,26),  
  Range = c(250,300,350,420,500),  
  Manufacturer = c("Tata","MG","Tesla","BYD","Tata")  
)  
  
ggplot(ev, aes(x=Battery, y=Price, size=Range, color=Manufacturer)) +  
  geom_point()
```

OUTPUT:

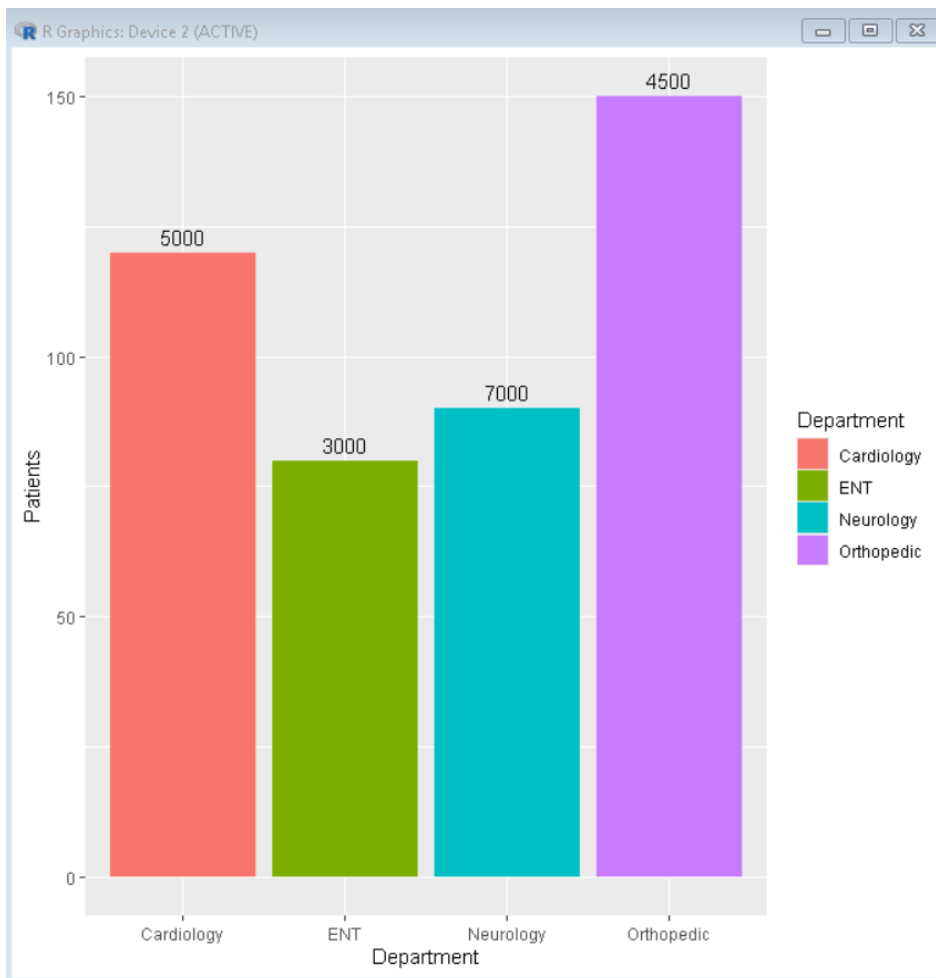


QUESTION 3: HOSPITAL PATIENT ADMISSION ANALYSIS

CODE:

```
hospital <- data.frame(  
  Department = c("Cardiology","Neurology","Orthopedic","ENT"),  
  Patients = c(120,90,150,80),  
  Cost = c(5000,7000,4500,3000)  
)  
  
ggplot(hospital, aes(x=Department, y=Patients, fill=Department)) +  
  geom_bar(stat="identity") +  
  geom_text(aes(label=Cost), vjust=-0.5)
```

OUTPUT:

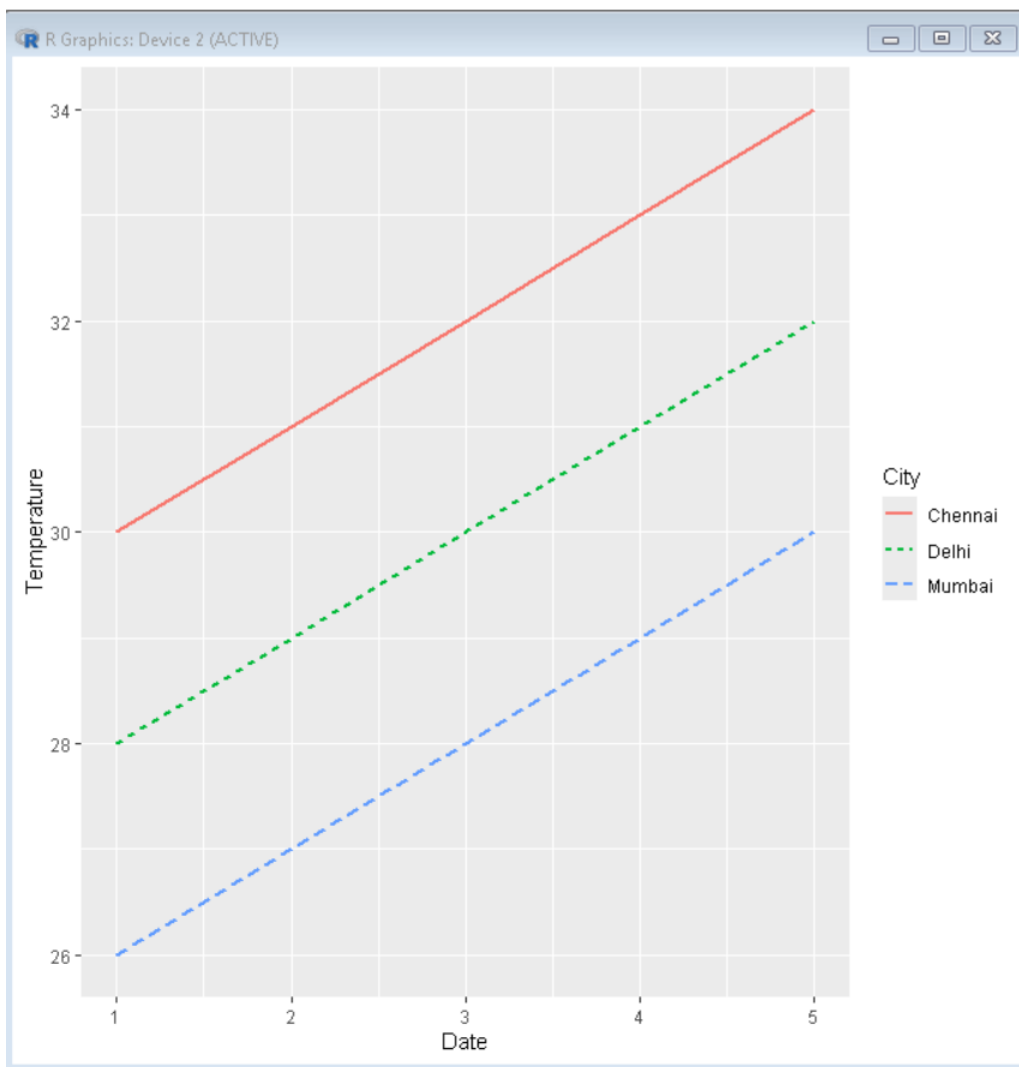


QUESTION 4: WEATHER MONITORING SYSTEM

CODE:

```
weather <- data.frame(  
  Date = rep(1:5,3),  
  Temperature = c(30,31,32,33,34,  
                  28,29,30,31,32,  
                  26,27,28,29,30),  
  City = rep(c("Chennai","Delhi","Mumbai"), each=5)  
)  
  
ggplot(weather,  
  aes(x=Date, y=Temperature,  
      color=City,  
      linetype=City)) +  
  geom_line(linewidth=1)
```

OUTPUT:

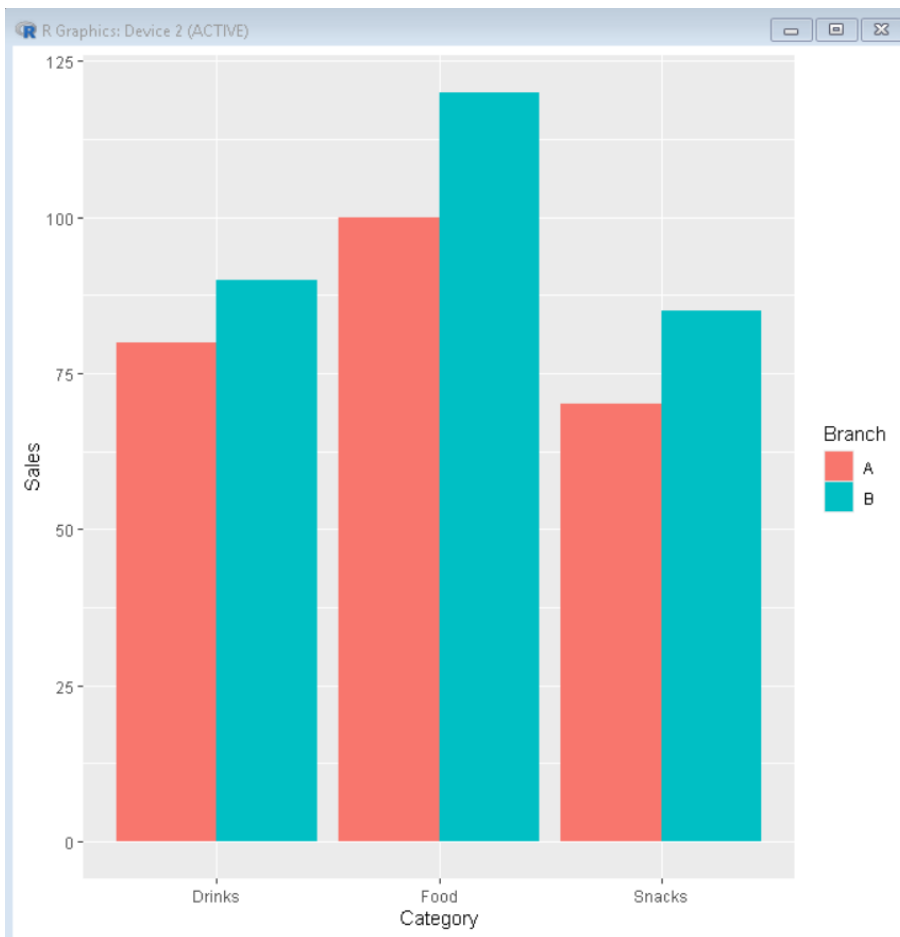


QUESTION 5: SUPERMARKET SALES ANALYSIS

CODE:

```
sales <- data.frame(  
  Category = c("Food","Food","Drinks","Drinks","Snacks","Snacks"),  
  Sales = c(100,120,80,90,70,85),  
  Branch = c("A","B","A","B","A","B")  
)  
  
ggplot(sales,  
  aes(x=Category,  
    y=Sales,  
    fill=Branch)) +  
  geom_bar(stat="identity", position="dodge")
```

OUTPUT:

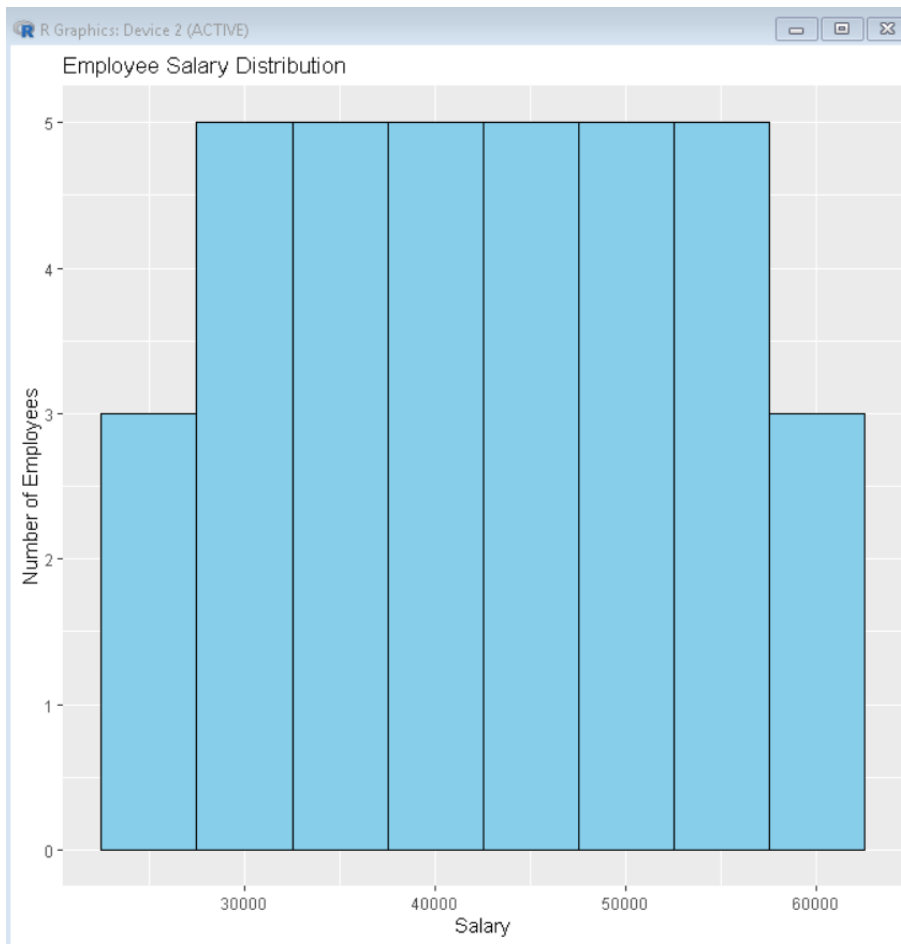


QUESTION 6: EMPLOYEE SALARY DISTRIBUTION

CODE:

```
salary <- data.frame(  
  Salary = c(  
    25000,26000,27000,28000,29000,  
    30000,31000,32000,33000,34000,  
    35000,36000,37000,38000,39000,  
    40000,41000,42000,43000,44000,  
    45000,46000,47000,48000,49000,  
    50000,51000,52000,53000,54000,  
    55000,56000,57000,58000,59000,60000  
  )  
)  
ggplot(salary, aes(x = Salary)) +  
  geom_histogram(binwidth = 5000, fill = "skyblue", color = "black") +  
  labs(  
    title = "Employee Salary Distribution",  
    x = "Salary",  
    y = "Number of Employees"  
  )
```

OUTPUT:

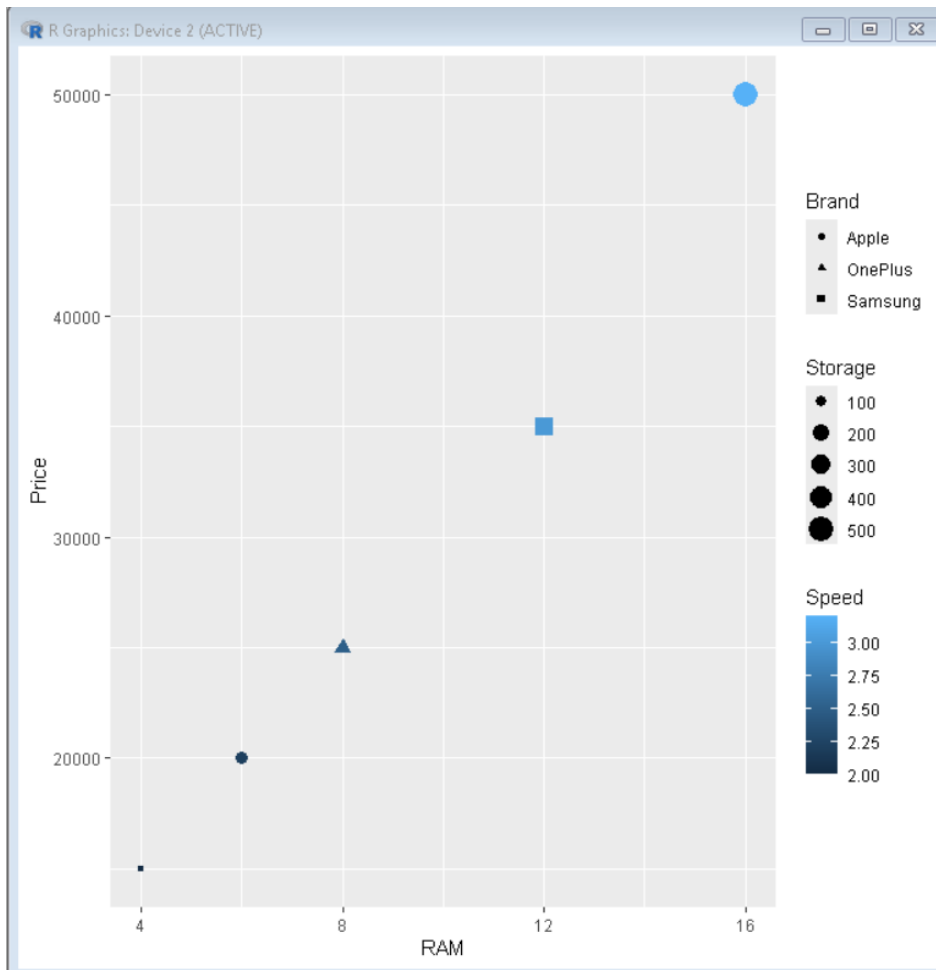


QUESTION 7: SMARTPHONE MARKET COMPARISON

CODE:

```
phone <- data.frame(  
  RAM = c(4,6,8,12,16),  
  Price = c(15000,20000,25000,35000,50000),  
  Brand = c("Samsung","Apple","OnePlus","Samsung","Apple"),  
  Storage = c(64,128,128,256,512),  
  Speed = c(2.0,2.2,2.5,3.0,3.2)  
)  
  
ggplot(phone,  
  aes(x=RAM,  
    y=Price,  
    shape=Brand,  
    size=Storage,  
    color=Speed)) +  
  geom_point()
```

OUTPUT:



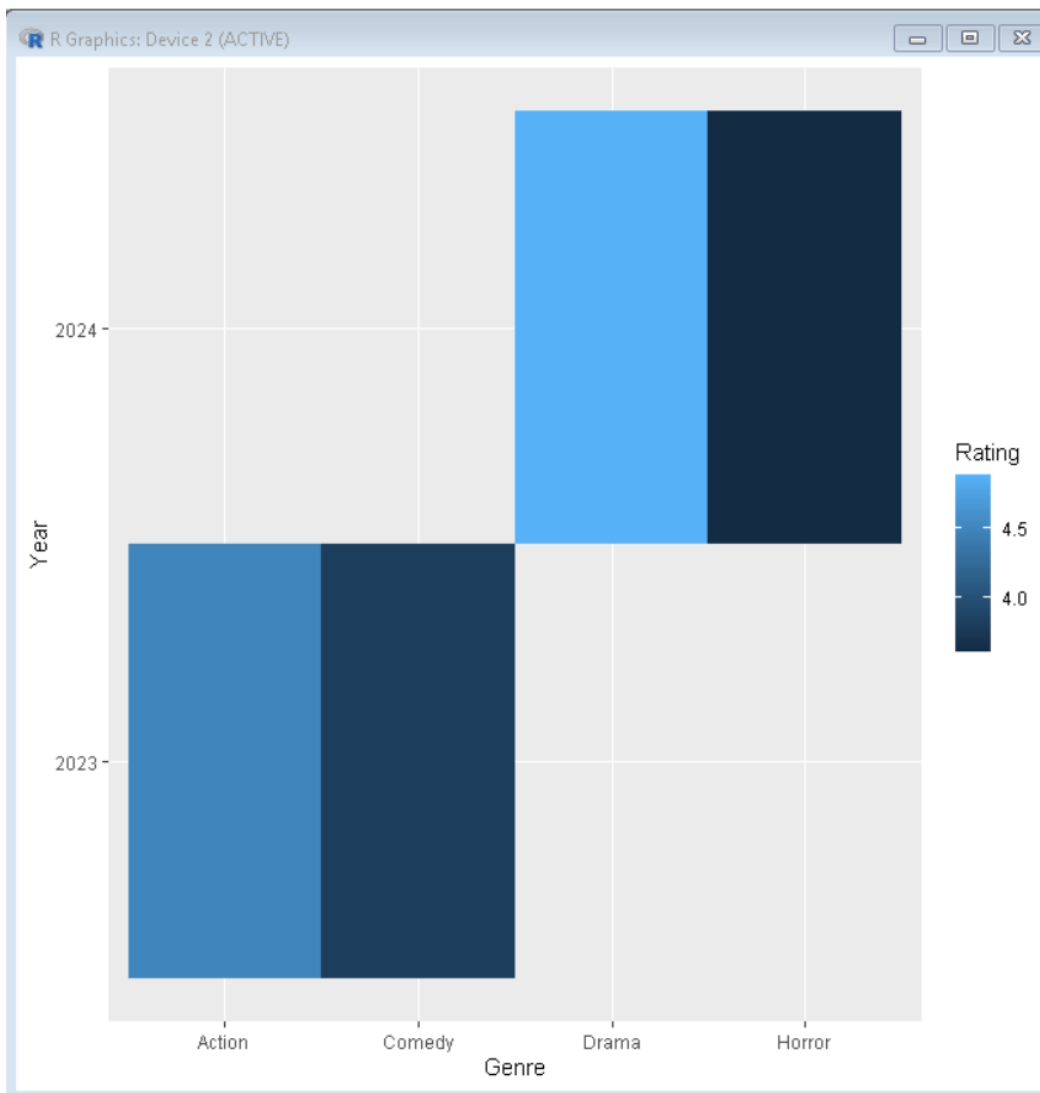
QUESTION 8: SMARTPHONE MARKET COMPARISON

CODE:

```
movies <- data.frame(  
  Genre = c("Action","Comedy","Drama","Horror"),  
  Year = c("2023","2023","2024","2024"),  
  Rating = c(4.5,3.8,4.9,3.6)  
)
```

```
ggplot(movies,  
  aes(x=Genre,  
      y=Year,  
      fill=Rating)) +  
  geom_tile()
```

OUTPUT:



QUESTION 9: COVID-19 STATE-WISE ANALYSIS

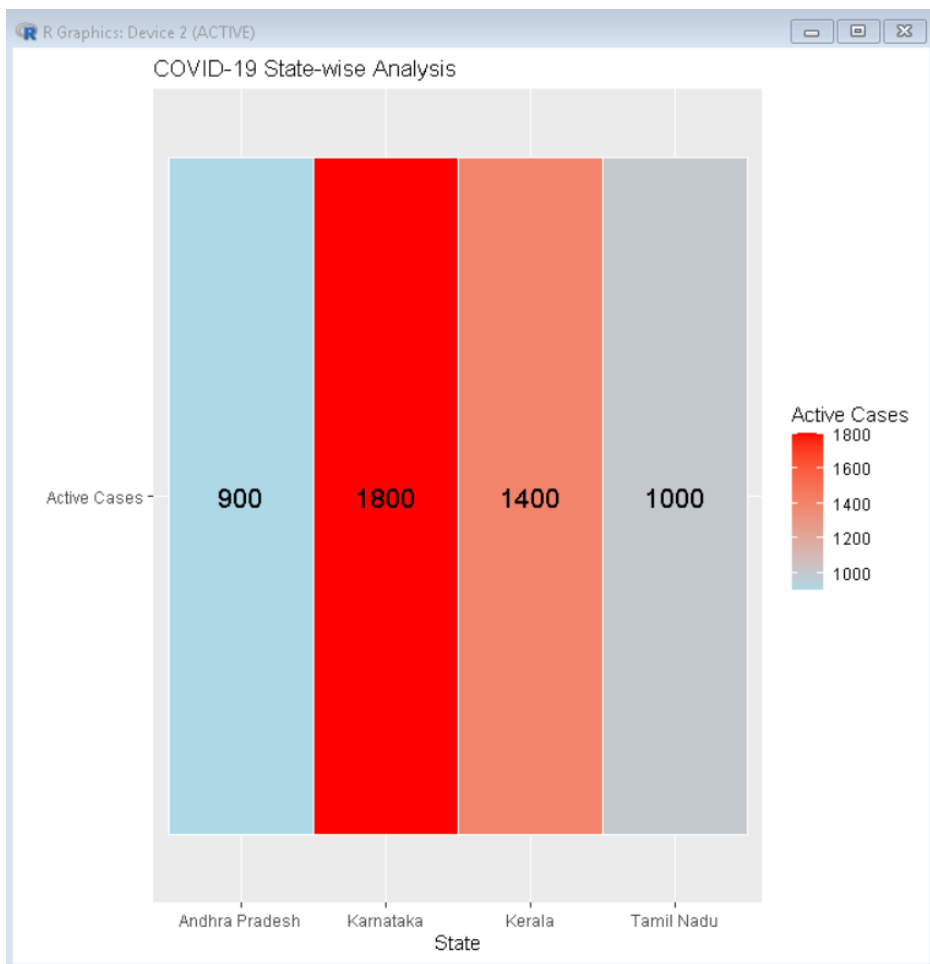
CODE:

```
library(ggplot2)

covid <- data.frame(
  State = c("Tamil Nadu", "Karnataka", "Kerala", "Andhra Pradesh"),
  Type = "Active Cases",
  Cases = c(1000, 1800, 1400, 900)
)

ggplot(covid, aes(x = State, y = Type, fill = Cases)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Cases), color = "black", size = 5) +
  scale_fill_gradient(low = "lightblue", high = "red") +
  labs(
    title = "COVID-19 State-wise Analysis",
    x = "State",
    y = "",
    fill = "Active Cases"
  )
)
```

OUTPUT:



QUESTION 10: IRIS FLOWER CLASSIFICATION

Code:

```
data(iris)

ggplot(iris,
  aes(x=Sepal.Length,
    y=Petal.Length,
    color=Species,
    shape=Species,
    size=Sepal.Width)) +
  geom_point()
```

OUTPUT:

